

Jayasri Suresh Vani

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EXECUTIVE SUMMARY

AI/ML Engineer with industry experience designing and deploying end-to-end machine learning systems, including LLM-powered applications, automated evaluation pipelines, and production ML workflows. Expertise in building scalable ML infrastructure, fine-tuning foundation models, and implementing robust model serving architectures with RESTful APIs. Proficient in Python, deep learning frameworks (PyTorch, TensorFlow, Keras), and MLOps practices including experiment tracking, model versioning, and CI/CD for ML. Strong foundation in NLP, computer vision, and statistical modeling, with proven ability to optimize model performance and deliver measurable business impact through data-driven solutions.

EDUCATION

Purdue University

Master of Science in Computer Science (GPA: 3.9/4.0)

Anna University - Easwari Engineering College

Bachelor of Technology in Information Technology (GPA: 4.0/4.0)

Fort Wayne, Indiana

Aug 2024 - May 2026

Tamil Nadu, India

Sep 2020 - May 2024

SKILLS

- **Programming Languages:** Python, Java, TypeScript, JavaScript
- **AI/ML & Data Science:** Deep Learning, NLP, Predictive Modeling, Computer Vision, Transformers, Sequence Models, Scikit-learn, TensorFlow, PyTorch, RAG, LangChain
- **Frameworks:** React, Next.js, Angular, NestJS, Flask, Django
- **Databases & Cloud:** MySQL, PostgreSQL, MongoDB, Firebase, Supabase, Docker, Prisma, FAISS
- **Tools:** Git, GitHub, REST APIs, MediaPipe

WORK EXPERIENCE

AI/ML Research & Development Intern | RAVEN AI | July 2025 - Present | New York, NY

- Advanced medical training evaluation via NLP-driven behavioral scoring across 5 communication dimensions.
- Benchmarked LLM-based evaluation pipelines using GPT-4o, Qwen3, and Llama3, applying prompt engineering and automated evaluation metrics (MAE, MSE, R²).
- Generated and validated 250+ synthetic doctor-patient conversations using contextual filtering and patient-name injection to expand dataset coverage.

Technical Support Engineer | PURDUE UNIVERSITY | Dec 2024 - Present | Fort Wayne, IN

- Resolved 200+ IT support tickets for network connectivity, hardware failures, and system access across university departments.
- Maintained 100+ university systems, ensuring high availability and minimizing downtime.
- Collaborated with IT teams to troubleshoot and resolve escalated technical issues.

Data Analyst & ML Intern | TECHVOLT PVT. LTD | Feb 2023 - Nov 2023 | Coimbatore, India

- Conducted exploratory data analysis (EDA) using Pandas, NumPy, and Seaborn to identify trends, anomalies, and actionable insights.
- Developed and evaluated diagnostic machine learning models including Logistic Regression, Naive Bayes, and Decision Trees, achieving 85%+ accuracy using Scikit-learn pipelines and cross-validation.
- Applied transfer learning by fine-tuning pre-trained models to improve predictive performance, incorporating data augmentation techniques and advanced visualizations to enhance model robustness and interpretability.

Software Developer Intern | OMEGA BINARIES | Apr 2023 - May 2023 | Chennai, India

- Built Java Swing biometric authentication system using MVC architecture, reducing latency by 40% and improving responsiveness by 25%.
- Integrated MySQL with AES-256 encrypted credential storage for secure authentication across 100+ users.
- Integrated MFS100 SDK for biometric verification and role-based access control.

PROJECTS

LumenDoc QA

/Jayasri2021/Document-QA-RAG

- RAG Q&A project that ingests PDFs or web URLs, splits them into chunks, and retrieves the most relevant passages to answer questions accurately using LangChain, FAISS, and OpenAI models.

Gesture2Globe- ALS sign detection

Jayasri2021/Gesture2Globe-ASL-Sign-Detection

- Built an end-to-end ASL recognition system with MediaPipe hand keypoints, TGCN ONNX inference, and a React+Flask pipeline that forms sentences (FLAN-T5), Google translate, and generates bilingual speech via PiperTTS from video and live webcam input.

MoodTunes -<https://moodtunes-423d2.web.app>

Jayasri2021/Music-Playlist-Gneneration-model

- Built a full-stack ML app classifying user emotions with 87% accuracy to generate personalized playlists; deployed on GCP Cloud Run

with secure authentication and API integration, and hosted a Logistic Regression model on HuggingFace Space using Gradio SDK.

NewsBias – Bias Detection and Controlled Text Rewriting

Jayasri2021/EAI_News_Media_Bias

- Implemented transformer-based NLP pipeline achieving 92.78% bias detection accuracy along with controlled rewriting using T5-based models with ROUGE-1 score of 0.6053 and explainability using LIME and SHAP.

Epileptic Seizure Classification | Python, CNN-LSTM, Signal Processing

Jayasri2021/Epileptic-Seizure-Classification

- Built a deep learning pipeline for EEG-based seizure detection using wavelet transforms and CNN-LSTM architectures.
- Achieved 92% accuracy with optimized preprocessing and model tuning.

CERTIFICATIONS

- **Google Cloud Badge:** *Introduction to Generative AI*
- **Kaggle:** *Into to Machine Learning, Intermediate Machine Learning*
- **Udemy:** *Flutter & Dart, Django4 & Python Fullstack Developer*